



Dietetics Malnutrition Diagnosis and Documentation

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	Dietetics & Nutrition	Dietitians

2. Purpose

Hospital acquired malnutrition (HAM) is diagnosed under the Hospital Acquired Complication, 13. Endocrine Complications as per the National Safety and Quality Health Service Standards and must be documented and coded accordingly. The purpose of this procedure is to ensure all patients who are malnourished or at risk of malnutrition are assessed appropriately and malnutrition diagnoses are clearly documented in the patient's medical record.

3. Procedure

Formal malnutrition assessment should be conducted on initial dietetic assessment, or within 3 working days of initial assessment for all patients:

- who were referred via the Malnutrition Screening Tool (MST),
- or have been identified at risk of malnutrition due to an acute/chronic condition.

3.1 Diagnosing Malnutrition in Adults

There are two ways to diagnose malnutrition in adults:

- Subjective Global Assessment (SGA) – gold standard/preferred
 - Dietitians to use the [SGA tool AKMR 60.3.5](#) on the Hub.
 - The SGA categorises the patient's malnutrition score as per the following:
A: No sign of malnutrition
B: Mild to moderate malnutrition
C: Severe malnutrition
- ICD-10 criteria - can be used where conducting an SGA is challenging or inappropriate (see Table 1 below).

Table 1: Criteria and corresponding ICD-10 codes regarding malnutrition diagnosis and documentation in adults.

Nutrition diagnosis	Documentation required
Severe protein energy malnutrition	• SGA = C OR • BMI <18.5 kg/m² <i>OR</i> unintentional loss of weight >10%, with • Evidence of suboptimal intake resulting in severe loss of subcutaneous fat and/or severe muscle wasting

Nutrition diagnosis	Documentation required
Moderate protein-energy malnutrition	- SGA = B OR - BMI <18.5 kg/m² <i>OR</i> unintentional loss of weight 5-9%, with - Evidence of suboptimal intake resulting in moderate loss of subcutaneous fat and/or moderate muscle wasting
Mild protein energy malnutrition	- SGA = B OR - BMI <18.5 kg/m² <i>OR</i> unintentional loss of weight 5-9%, with - Evidence of suboptimal intake resulting in mild loss of subcutaneous fat and/or mild muscle wasting
Unspecified protein energy malnutrition	- MST ≥ 2 AND - Unintentional weight loss (>10% indefinite time or >5% in last three months) <i>OR</i> - BMI <20kg/m² (if under 70 years of age) or BMI <22kg/m² (if over 70 years of age) WITH - Evidence of suboptimal intake resulting in loss of subcutaneous fat and/or muscle wasting

It can be challenging to assess a patient's nutritional status when:

- unable to obtain patient's weight, weight history or BMI
- the patient is fluid overloaded
- you are unable to complete an SGA

Using an evidence-based nutrition-focussed physical exam (NFPE) when interviewing the patient can aid nutritional assessment (see [Appendix 1](#)).

3.2 Diagnosing Malnutrition in Paediatrics

Formal malnutrition assessment should be conducted on initial dietetics assessment, or within 3 working days of initial assessment for all patients:

- who were referred via the Paediatric Nutrition Screening Tool (PNST)
- or have been identified at risk on daily screen for inadequate nutrition using the AKG Paediatric Skin Assessment, Pressure Injury and Falls Risk Assessment & Management form.

There are two ways to diagnose malnutrition in paediatrics:

- Complete weight, height or length measurements. Measurement of head circumference is completed at birth and 6-8 weeks of age.
 - Weight is measured with children less than 1 year of age being bare, and children over 1 year wearing only a dry nappy or light clothing. Document what the child is wearing at the time of measurement.

- Height or length measurement is completed with a stadiometer or a length board/roll.
- Measurements of lower leg length or knee-heel length can be used for growth if accurate height or length cannot be obtained using standard methods. Height is estimated from lower leg length using the Chumlea equation for age.
- Plot measurements on the growth chart in the medical record.
 - Make corrections for prematurity for infants and children aged 0-2 years.

Table 2: Criteria and corresponding ICD-10 codes regarding malnutrition diagnosis and documentation in paediatrics.

Nutrition diagnosis	Documentation required
Mild protein energy malnutrition	0-2 years age • Weight for length from World Health Organisation (WHO) growth charts -1 to -1.9 z-score • Presence of stunting
	2-20 years • BMI from Centres for Disease Control (CDC) growth charts -1 to -1.9 z-score • Unintentional weight loss 5-7.4% • Presence of stunting
Moderate protein energy malnutrition	0-2 years age • Weight for length from WHO growth charts -2 to -2.9 z-score, moderately wasted • Presence of stunting
	2-20 years • BMI from CDC growth charts -2 to -2.9 z-score, moderately wasted • Unintentional weight loss 7.5-9.9% • Presence of stunting
Severe protein energy malnutrition	0-2 years age • Weight for length from WHO growth charts <-3 z-score, severely wasted • Presence of stunting
	2-20 years • BMI from CDC growth charts <-3 z-score, severely wasted • Unintentional weight loss >10% • Presence of stunting

4. Documentation

Episodes of care such as acute and rehabilitation or acute and mental health are classified as separate admissions. It is essential that if HAM is identified, it is documented by a dietitian, in

both episodes of care. For example, if a dietitian diagnoses and documents HAM in the medical record on the medical ward, when the patient transfers to the rehabilitation ward and malnutrition is still present, the rehabilitation dietitian must document HAM again to ensure identified for both admissions.

Documentation is to be completed in the patients' medical record using the ADIME (assessment, diagnosis, intervention, monitoring, evaluation) format following the nutrition care process. Documentation should include:

4.1.1 Assessment

- Intake: suboptimal intake compared to usual intake and/or estimated energy requirements.
- Anthropometric measures: weight, weight history and percent weight loss, SGA score/category referencing medical record number (AKMR60.3.5).
- Nutrition focused physical findings: muscle wasting and subcutaneous fat wasting.
- Malnutrition Diagnosis: Copy and paste the digital medical record (DMR) Malnutrition Diagnosis Template into DMR and delete irrelevant terminology. Please see [Appendix 2](#) or W:\Dietetics & Nutrition\AKH\Clinical Resources\DMR\Templates. The template includes:
 - The **degree** of malnutrition (mild, moderate, severe OR unspecified). If the degree of malnutrition changes throughout the patient's admission, it must be documented again in the medical record. The most severe degree of malnutrition during that admission will be coded.
 - The **onset of malnutrition**. If the patient was malnourished prior to admission (pre-existing) OR developed malnutrition during their hospital admission (hospital acquired):
 - Repeated/weekly MST can help guide when malnutrition developed.
 - If hospital acquired malnutrition is diagnosed, the dietitian is to investigate whether it could have been prevented (i.e. with earlier dietetic intervention if delayed referral). If was deemed as preventable, complete a Datix Clinical Incident Management (CIM) form.
 - Pre-existing malnutrition that worsens over the hospital admission is not considered hospital acquired and should be documented as pre-existing.
 - Indicate the **aetiology** of malnutrition. Is it from acute/ injury, a chronic disease or from environmental reasons.
 - Indicate **dietetic assessment** used to diagnose the malnutrition (SGA or ICD-10)

4.1.2 Diagnosis

- Problem: although malnutrition cannot often be acutely resolved, it is recommended that it be selected as the primary diagnosis to highlight that a patient is malnourished.
- Aetiology: factors gathered during nutrition assessment which contribute to the existence of malnutrition.
- Signs/symptoms: SGA score, percentage weight loss, muscle and/or fat wasting and estimated intake.

4.1.3 Intervention

- Nutrition goal and prescription, education and coordination of nutrition care with multidisciplinary team.

4.1.4 Monitoring and Evaluation

- Aetiology of malnutrition and other parameters related to the patient's dietetic intervention.

5. Roles and responsibilities

Role	Responsibility
Dietitian	Assess, diagnose and document malnutrition, including severity and hospital acquired or pre-existing. As requested by coding or Coordinator of Dietetics post audit, investigate appropriate malnutrition diagnosis in a patients' medical record and add a new entry if correction is required.

6. Supporting information

The following documents support the implementation of this procedure:

- AKG Subjective Global Assessment AKMR60.3.5

7. Compliance, monitoring and evaluation

Compliance against this procedure will be evaluated via medical record documentation audits and monitoring hospital acquired malnutrition data by the Nutrition and Dietetics Department.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.10: Screening of risk:
- Standard 5 – Action 5.11: Clinical Assessment
- Standard 5 – Action 5.12: Developing the comprehensive care plan
- Standard 5 – Action 5.28: Nutrition and Hydration

9. References

1. Walton, KL. Treating malnutrition in hospitals: dietitians in the driving seat, Nutrition & Dietetics, 66(4), 2009, 202-205.
2. Ferguson, M et al. Development of a validated and reliable malnutrition screening tool for adult acute hospital patients, Nutrition, 15 (6), 458-464
3. Detsky, A.S et al. What is subjective global assessment of nutritional status, Journ Par & Ent Nut, 11 (1), 1987, 8-13.
4. Academy of Nutrition and Dietetics. Nutrition terminology reference manual (eNCPT): Dietetics Language for Nutrition Care. <http://ncptwebauthor.com>. Accessed 17 December 2015.

5. Mauldin Ketal. Performing nutrition assessment remotely via telehealth, Nur Clin Pract. <http://pubmed.ncbi.nlm.nih.gov/34101249/>. Accessed 2 March 2021.

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
Malnutrition	An inadequate intake of protein or energy over time resulting in impaired growth and development in children and a loss of fat and muscle stores in adults. Malnutrition can occur at any body mass index (BMI).
Hospital acquired complication	A hospital-acquired complication refers to a complication for which clinical risk mitigation strategies may reduce (but not necessarily eliminate) the risk of that complication occurring.
Hospital acquired malnutrition	Malnutrition that develops during a hospital admission.

11. Policy owner

Coordinator Nutrition & Dietetics

Enquiries relating to this procedure may be directed to:

Title: Coordinator Nutrition & Dietetics
 Department: Dietetics & Nutrition Department
 Email: ahs.dietetics@health.wa.gov.au

12. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	13/11/2024	13/11/2027	Throughout document – reflecting change to DMR Section 3 – Expanding on nutrition diagnosis Table 1 – expanding on nutrition diagnosis criteris

13. Authorisation

This procedure has been authorised and issued by the Director Allied Health and Ambulatory Care.

Authorised by	Danielle Kilmurray – Director of Allied Health and Ambulatory Care
Authorisation date	24/06/2024
Published date	13/11/2024

Appendix 1 - Evidence based nutrition-focussed physical exam (NFPE)

NFPE Component	Provider Assessment Actions	Assessment Questions to Ask the Patient
Functional Status	- Observe posture/position/ physique - Notice if the patient is sitting or lying in bed - Notice if the patient is able to maintain posture - Notice if the patient is in a wheelchair and/or requires a walker	- Do you notice signs of weakness? Could these be related to changes in nutrition? - Has your activity level changed? If so, how? (address activities of daily living and time spent on physical activities)
Fat and Muscle Wasting	- Notice if the patient appears to be underweight or overweight - Notice visible losses of fat or muscle - Guide patients to show each area of the body - Patients may have to step away so dietitian can see larger area (clavicle, scapular, thoracic/ lumbar, anterior thigh) - Explain positioning (bend elbow at right angle, pinch index finger and thumb) - Consider that oedma can mask weight changes - For monitoring and follow-up evaluation purposes, consider asking for photos	- Does your face appear thinner to you? - Suggest patients look back at recent photos for changes - If applicable, ask whether dentures fit the same - Do your glasses or jewellery fit differently? - Have you noticed any changes in weight? - Do your clothes fit differently? - Are you using your belt on a tighter notch? - Does your appearance look different to you in any way? (Specify area)
Micronutrients: Skin	- Observe facial skin while speaking to patient - Notice any discolouration, rashes, or poor pallor	- Have you noticed any rashes or changes on your skin, areas of flaky skin, redness, or wounds? - When did these appear?
Micronutrients: Hair	- Observe hair while speaking to patient - Ask to see arm hair	- Have you noticed any difference in your hair? - Has your hair been falling out more? Are you waking up with more hair on your pillow? - Any changes in arm hair or does arm hair look like a corkscrew?

NFPE Component	Provider Assessment Actions	Assessment Questions to Ask the Patient
Micronutrients: Nails	- Ask to see nails close up - Ask patient to push on nail bed to demonstrate blanching	- Have you noticed any difference with your nails? - Are you nails breaking more easily?
Micronutrients: Eyes	- Observe eyes while speaking to patient - Ask patient to lightly pull down on bottom eyelid to assess conjunctiva colour	- Have you noticed any changes in eye health or colour? - Have you had any trouble seeing in general or seeing at night? - Are you experiencing dry, stinging or burning eyes?
Micronutrients: Mouth	- Observe mouth and lips - Ask to see tongue, inner bottom lip, and gums*	- Have you noticed any sores in your mouth or changes with your teeth? - Are you having any difficulty chewing or swallowing? - Any trouble with bleeding gums? - Any taste changes? - Any burning sensations in your mouth?
Fluid Status	- Teach patient to assess quality of oedema - Teach patient skin turgor assessment (back of hand, forearm, sternum)	- Have you noticed any increases in puffiness/fluid accumulation in your ankles/around your feet? - How far up your leg does the swelling occur? - Is there an indentation when you press on your skin? - How long does the indentation last? - Pinch skin, after letting go, does the skin return to its original position? How long does it take?

*Also, can be used to assess fluid status

Appendix 2 –DMR Malnutrition Diagnosis Template

Delete what is not applicable. Only choose ONE for each category (most appropriate).

MALNUTRITION

Degree

Mild

Moderate

Severe

Unspecified protein energy malnutrition

Onset

Pre-existing

Hospital acquired

Aetiology

Acute/injury

Chronic disease

Environmental (starvation)

Dietetic Assessment

Subjective global assessment (AKMR6.03.5)

ICD-10 criteria

Example of completed Malnutrition DMR template:

MALNUTRITION

Degree

Moderate

Onset

Pre-existing

Aetiology

Chronic disease

Dietetic Assessment

ICD-10 criteria